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Capacity Assignment in Diffserv Networks for Proportional QoS

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Abstract

The Differentiated Services (Diffserv) architecture is a scalable solution for providing Quality of Service (QoS) over packet switched networks. Yet, Diffserv cannot offer strict performance guarantees to its subscribers. We consider in this paper an alternative notion of *relative* performance guarantees. Specifically, the network manager's goal is to maintain pre-defined ratios between certain congestion measures over the different service classes. For that purpose, the manager may judiciously allocate available capacity between the service classes. This scheme is studied within a flow model, with self-optimizing users, which are subject to static SLAs. We pose the entire problem as a *non-cooperative game*. Starting with a simplified network model, we demonstrate the existence and uniqueness of the equilibrium, and show how to calculate the required capacity assignment. We then extend these results to a general network topology. Finally, the incorporation of dynamic SLAs is briefly addressed.

1 Introduction

1.1 Background and Motivation

The need for providing service differentiation over the Internet is an ongoing concern in the networking community. The Differentiated Services (Diffserv) architecture [5] has been proposed by the IETF as a scalable solution for QoS provisioning. Instead of reserving resources per session (e.g., as in the Integrated Services (IntServ) model [24]), packets are marked to create a smaller number of packet classes, which offer different service qualities. The Diffserv proposal suggests to combine simple priority mechanisms at the network core with admission control mechanisms at the network edges only, in order to create diverse end-to-end services.

The two principal Diffserv classes that have been formalized are the Expedited Forwarding (EF) [9] and the Assured Forwarding (AF) [14] services. The premise of the EF is to provide no-loss and delay reduction to its subscribers. AF is intended for users who need reliable forwarding even in times of network congestion. A Service Level Agreement (SLA) is formed between the user and the provider, in which the user commits to interact with the network in a given way, usually reflected by the allowed bandwidth for each service class. Ongoing IETF work concentrates on defining the engineering and architectural aspects of Diffserv-enabled routers (e.g., [3]). However, current technical specifications deliberately do not quantify the provider part of the agreement, i.e., the actual *service characteristics*, which users will obtain by using the above mentioned classes. Hence, the two elements that jointly determine the QoS in the different service classes, namely, the resource allocation policy (which may be carried out by internal packet scheduling rules) and the regularization of user traffic (e.g., by admission control or pricing) are left as open design issues. Apparently, service characteristics would have to be defined and publicly declared in order to make the distinction between the service classes meaningful to the user (and possibly worth paying for).

The Diffserv network cannot offer strict quality guarantees, as resources are allocated to the service classes based on some average network conditions [6]. Instead, the provider may declare upper-bounds on QoS measures, or alternatively provide looser guarantees, such as probabilistic or time-dependent guarantees. Another option is to offer *relative* quality guarantees. This option can be easily quantified and advertised, as illustrated by the following

exemplifying rule: "service class Q will offer an end-to-end average delay, which is at least two times less than any other class, independent of the level of congestion". When a user signs an SLA for class Q it is aware of what it gets, and expects the provider to uphold the agreement conditions. In this paper, we focus on the *proportional* QoS model, whereby QoS ratios between the classes are announced. Specifically, we concentrate on *delay* ratios, although analogous definitions may be suggested when considering other QoS measures. The proportional QoS model benefits from implementation-related pros. Ratios are easier to maintain in comparison with absolute end-to-end guarantees, primarily because they may hold for different levels of congestion, and secondly because keeping the ratios locally (on a node basis) leads to fulfilling this objective on the network level.

We shall examine capacity allocation as the main network management tool for achieving the proportional QoS design objective. The goal of capacity allocation is to keep the announced delay ratios, irrespectively of complementary means for network traffic control, such as pricing and admission control. Each network node owns a fixed amount of capacity to be divided among its active service classes. Generally, it is straightforward to calculate the capacity allocation when the traffic in each service class is fixed. In this paper, however, we consider the interaction of the user behavior and network conditions. Within a standard flow model (see [4], Sec. 5.4) we represent the user population as a (finite, heterogenous) set of self-optimizing decision makers. The users can shift their traffic from one service class to the other in response to current congestion conditions, subject to their SLAs. The Diffserv provider assigns a *capacity manager* in order to maintain the delay-ratios design objective, in face of changing network conditions. We pose the overall situation as a non-cooperative game between the manager and the (selfish) users, and explore the associated capacity management policies and equilibrium conditions.

1.2 Related Literature

Service differentiation approaches: The simplest approach for providing differentiated services is Odlyzko's Paris Metro Pricing (PMP) proposal [19]. The idea of PMP is to partition a single physical network into several logically separated channels, each of which treats all packets equally. The only difference between the channels is their price, therefore channels with higher prices would attract less traffic and provide better service. Simple mathematical models (see e.g., [6, 19]) suggest that the above separation, along with appropriate pricing

of the service classes, may yield a better social welfare, comparing to the single channel case.

Other papers explicitly consider the following elements, in addition to price-based differentiation:

- *The user model* – the assumptions on the users preferences, their size etc.
- *The scheduling mechanism* which is used for service discrimination. Common choices are priority queues (e.g., [12, 18, 17]) and the GPS scheduler [13].
- *The network objective.* The most common objectives are social welfare (e.g., [12, 18]), i.e., maximizing the sum of user utilities, and profit maximization (e.g., [13, 17]).

A major concern of these (and similar) service differentiation approaches is in calculating the prices that would lead to achieving the network objective. Certain assumptions are made on the user model in order to calculate these prices. Mostly, some parametric or probabilistic description is used to represent the heterogonous sensitivity to QoS. An additional common ground is that the network does not declare any kind of QoS guarantees. Users are assumed to acquire the best deal there is with respect to their quality-price tradeoff.

Selfish routing models: Game-theoretic analysis is widely used to study network working conditions. The involved issues include the existence and uniqueness of an equilibrium point, its calculation, and its properties (such as the degradation of performance due to selfish behavior, known as the "price of anarchy" [23]). A common routing model, originated from the field of transportation networks, has considered networks shared by infinitesimal users (e.g., [8], [7], and [1] for a survey). Incentive-compatible prices have been used for inducing the network to the socially optimal flows [7]. The latter approach has been exploited for providing QoS guarantees in communication networks [16]. The case of finitely many users, each carrying substantial flow has been introduced to the networking literature more recently (see [20], [2], [21], [15]). In [15], the equilibrium properties are applied for network design (namely, link capacity assignment), where the objective is to obtain the socially optimal working point. We use a similar routing model to represent the user's choice of service class.

Proportional QoS: Dovrolis et al. [10] (and references therein) proposed a class of schedulers, based on the Proportional Delay Differentiation (PDD) model. This model aims at providing predetermined delay ratios. The schedulers are implemented by observing the history of the encountered delays (or alternatively, by measuring the delay of the packet at the

head of each service class), and serving the class which most deviates from its nominal delay ratio. In the present work we do not rely on PDD schedulers, but rather use a capacitated links model that may be considered a proxy to existing scheduling schemes such as GPS.

1.3 Contribution and Organization

In this paper, we propose schemes for inducing proportional QoS through capacity allocation, when users can react to the allocation decisions. Thus, the schemes are adaptive. Our starting point is the single link network, in which each service-class is modelled as a separate arc. The precise definitions of the network and user models are given in Section 2. In Section 3 we show that for every given load and required delay ratios, there exists a unique equilibrium point, in which the declared ratios are fulfilled. Furthermore, this equilibrium may be efficiently calculated. These results serve as the theoretical ground for capacity updating schemes that allow the network reach the appropriate equilibrium. The possession of user-specific information, usually the case at edge routers, leads to an explicit calculation of the capacities. In an alternative reactive scheme, more suitable for core routers, the network adapts its capacity based on class information only. In Section 4 we extend our model to general network topologies. We show that the principles that were developed in Section 3 can be carried to the general topology case. Section 5 discusses the incorporation of dynamic SLAs and pricing, and is followed by some concluding remarks.

2 The Single Link Model

The single link model assumes that all users use the same link for shipping their flow. As a stand-alone model, it may be viewed as an approximation of a single path in a network, where the variations in traffic due to other (intersecting) network paths are neglected. More importantly, our results for this simplified model will serve as the building blocks when considering the general network case (Section 4).

We consider a set $\mathcal{I} = \{1, 2, \ldots, I\}$ of users, which share a single link that offers a set of service classes $\mathcal{A} = \{1, 2, \ldots, A\}$. Each user $i \in \mathcal{I}$ has a throughput demand, which is a constant r^i. A user ships its demand by splitting it among the service classes. Since each service class is characterized by its own performance measure (to be described in the sequel), it would be convenient to consider the link with its respective service classes as a two

terminal (source-destination) network, which is connected by a set of parallel arcs. Each arc represents a different service class. Thus, the set of arcs is also denoted by $\mathcal{A}$, and the terms service class and arc are used interchangeably (see Figure 1 for a network example). We denote by f_a^i the flow which user i ships on arc a. User i can choose the flows f_a^i, subject to the flow constraint $\sum_{a\in\mathcal{A}} f_a^i = r^i$. Each user is also characterized by its signed SLAs. These SLAs upper bound the flow which each user is allowed to ship into each of the different service classes. Denoting by s_a^i the upper flow bounds of user i, then $0 \le f_a^i \le s_a^i$ must hold for every $i \in \mathcal{I}$ and $a \in \mathcal{A}$. These SLAs, whose negotiation is exogenous to our model, reflect the user preference regarding the price vs. quality tradeoff (as SLAs should be charged in some way). Turning our attention to an arc $a \in \mathcal{A}$, let f_a be the total flow on that arc, i.e., $f_a = \sum_{i\in\mathcal{I}} f_a^i$; also, denote by $\mathbf{f}_a$ the vector of all user flows on arc a, i.e., $\mathbf{f}_a = (f_a^1, \ldots, f_a^I)$. The user flow configuration $\mathbf{f}^i$ is the vector $\mathbf{f}^i = (f_1^i, \ldots, f_A^i)$. The flow configuration $\mathbf{f}$ is the vector of all user flow configurations, $\mathbf{f} = (\mathbf{f}^1, \ldots, \mathbf{f}^I)$. A user flow configuration is *feasible* if its components obey the nonnegativity, the SLA, and the demand constraints, as described above. Denote by $\mathbf{F}^i$ the set of all feasible user flow configurations $\mathbf{f}^i$, and by $\mathbf{F}$ the set of all feasible flow configurations $\mathbf{f}$.

The network manager has a constant capacity C, which it decides how to divide between the service classes. We denote by c_a the allocated capacity at arc a. The capacity allocation of the manager is the vector $\mathbf{c} = (c_1, \ldots, c_A)$. An allocation $\mathbf{c}$ is feasible if its components obey the nonnegativity and total capacity constraint, namely (i) $c_a \ge 0$, $a \in \mathcal{A}$ and (ii) $\sum_{a\in\mathcal{A}} c_a = C$. The set of all feasible capacity allocations $\mathbf{c}$ is denoted by Γ. Finally, a system configuration is feasible if it is composed of a feasible flow configuration and a feasible capacity allocation.

The *performance measure* of both the users and the manager is specified through their respective cost functions, which they wish to minimize. We denote by $J^i(\mathbf{f}, \mathbf{c})$, $i \in \mathcal{I}$ the cost function of user i, and by $J^M(\mathbf{f}, \mathbf{c})$ the cost function of the manager. The costs of the users and the manager are related to the *congestion* level at each of the service classes. We shall use the well known *M/M/1 latency function*

$$D_a(f_a, c_a) = \begin{cases} \frac{1}{c_a - f_a} & f_a < c_a \\ \infty & \text{otherwise} \end{cases} \tag{1}$$

as a congestion measure of each service class. Here c_a is the transmission capacity measured in the same units as the flow f_a. This latency function is often used to model delay costs in

communication networks [4], and provides a clear sense of link capacity.

User cost functions. Each user i wishes to minimize its total delay, namely,

$$J^i(\mathbf{f}, \mathbf{c}) = \sum_{a=1}^{A} f_a^i D_a(f_a, c_a), \tag{2}$$

where D_a is defined in (1).

Manager cost function. The objective of the manager here is to impose certain predetermined ratios between the average delays of the service classes. Taking the delay of class 1 as a reference, the ratios are described by a vector $\rho = (\rho_2, \ldots, \rho_A)$, $0 < \rho_a < \infty$, where the manager's objective is to have the delays $D_1, \ldots, D_A$ obey

$$D_a(f_a, c_a) = \rho_a D_1(f_1, c_1), \quad \text{for } a = 2, \ldots, A. \tag{3}$$

We refer to that relation as the *fixed ratio objective*. In functional terms, the manager's cost function may thus be defined as

$$J^M(\mathbf{f}, \mathbf{c}) = \begin{cases} 0 & \text{if (3) holds,} \\ \infty & \text{otherwise.} \end{cases} \tag{4}$$

An alternative objective of the manager that will be considered, is to minimize a weighted sum of the delay functions, that is

$$\bar{J}^M(\mathbf{f}, \mathbf{c}) = \sum_{a \in \mathcal{A}} \beta_a D_a(f_a, c_a), \tag{5}$$

where $\beta_a > 0$, $a \in \mathcal{A}$. As we shall see, there is a close relation between the fixed ratios objective and the weighted sum objective (5).

The following feasibility condition will be assumed throughout our analysis of the single link model:

Assumption 1

(i) $r^i \leq \sum_{a=1}^{L} s_a^i$ *for every* $i \in \mathcal{I}$.

(ii) $\sum_{i=1}^{I} r^i < C$

The first part of the assumption ensures that each user has a feasible flow allocation within its SLA bounds; the second part ensures that the network has enough capacity to support user demands.

The interaction between the manager and the users will be referred to as the *users-manager game*. A Nash Equilibrium Point (NEP) of that game is a feasible system configuration $(\tilde{\mathbf{f}}, \tilde{\mathbf{c}})$ such that all costs (J^M, J^i $\forall i \in \mathcal{I}$) are finite, and the following conditions hold:

$$\begin{aligned} J^M(\tilde{\mathbf{f}}, \tilde{\mathbf{c}}) &= \min_{\mathbf{c}\in\Gamma} J^M(\tilde{\mathbf{f}}, \mathbf{c}), \\ J^i(\tilde{\mathbf{f}}^i, \tilde{\mathbf{f}}^{-i}, \tilde{\mathbf{c}}) &= \min_{\mathbf{f}^i\in\mathbf{F}^i} J^i(\mathbf{f}^i, \tilde{\mathbf{f}}^{-i}, \tilde{\mathbf{c}}) \quad \forall i \in \mathcal{I}, \end{aligned} \tag{6}$$

where $\tilde{\mathbf{f}}^{-i}$ stands for the flow configurations of all users, but the ith user. Namely, the NEP is a network working point, where no user, nor the manager, finds it beneficial to change its flow or capacity allocation.

3 Capacity Assignment and Equilibrium Analysis

In this section we present the analysis of the single link model. Section 3.1 presents our analytical main results, whose proofs are deferred to Section 3.3. Section 3.2 presents a detailed discussion of these results, and their implication for capacity management.

3.1 Main Results

In this section we analyze the equilibrium point and suggest capacity assignment schemes that induce the ratios objective. First, we show that the manager has a unique best response with respect to the ratio objective (3). This response is a simple solution to a set of linear equations. Then we prove the existence and uniqueness of an equilibrium point, in which the desired ratios are met (for every given ρ). Consequently, we show the equivalence between the best response with respect to (3) and the best response with respect to the modified manager objective function (5). The last fact is significant in terms of the preferred algorithmic solutions for adaptation. Finally, we show that the unique equilibrium can be *explicitly calculated* via solving quadratic optimization problems. This property indicates the possible use of a deliberative approach in setting the capacities.

Theorem 1 considers the best response capacity assignment of the manager, given any (fixed) flow configuration.

Theorem 1 *Consider a fixed flow configuration $(f_1, \ldots, f_A)$ and a desired ratio vector ρ. Then there exists a unique capacity allocation* $\mathbf{c}$ *such that (3) is met. This allocation is explicitly given by*

$$c_a - f_a = (C - \sum_{k \in \mathcal{A}} f_k) \frac{\rho_a^{-1}}{\sum_{k \in \mathcal{A}} \rho_k^{-1}}, \tag{7}$$

where $\rho_1 \stackrel{\triangle}{=} 1$.

We note that $\sum_{\alpha \in \mathcal{A}} f_\alpha = \sum_{i \in \mathcal{I}} r^i$. However, the first quantity is the one observed by the manager.

Theorem 2 *For every delay ratios vector ρ, there exists a unique Nash equilibrium point. This NEP has finite costs and the ρ ratios are met in this point.*

Theorem 3 *Consider the users-manager game, whose NEP is defined in (6), and an additional game, where J^M is replaced by $\bar{J}^M$ (accordingly, J^M is replaced by $\bar{J}^M$ in (6) for the NEP definition of the latter). If the parameters of J^M and $\bar{J}^M$ are such that $\beta_a = \frac{1}{\rho_a^2}$ for every $a = 1, \ldots, A$ where $\rho_1 \stackrel{\triangle}{=} 1$, then the two games are equivalent in the sense that their (unique) equilibrium points coincide.*

Theorem 4 *For every given ratio vector ρ, the capacity allocation at the (unique) NEP can be explicitly calculated by solving the following I quadratic optimization problems (with A variables each)*

$$\begin{cases} \min_{f_1^i, \ldots, f_A^i} \sum_{a=1}^{A} \frac{1}{2} D_a^2 {f_a^i}^2 + D_a f_a^i \\ \text{s.t} \qquad 0 \le f_a^i \le s_a^i \\ \qquad\quad \sum_{a=1}^{A} f_a^i = r^i, \end{cases}$$

where D_a, $a \in \mathcal{A}$ are the (known) delays at the equilibrium.

Theorem 5 *For every given* fixed *capacity allocation* $\mathbf{c}$*, the resulting equilibrium flow is unique.*

3.2 Discussion

We next discuss some consequences of the previous results, as well as some aspects of our modelling assumptions.

Theorem 4 provides the means for the explicit calculation of the capacity allocation at equilibrium, which induces the required delay ratios. Indeed, once the manager sets these capacities, all it needs to do is to wait for the user flows to settle into their respective equilibrium point. As these flows are unique (Theorem 5), the result will indeed be the desired NEP which satisfies the delay ratios. From a computational viewpoint, the equilibrium capacities are derived by solving I quadratic optimization problem with A variables each. Since the number of classes in a Diffserv network is small, the calculation procedure is computationally manageable.

On the down side, the explicit computation of the equilibrium capacities requires the manager to possess considerable per-user information, namely user's objective function, its SLAs and flow demand. While such information may be available for edge routers, this is certainly not the case for core routers, which are typically aware only of local per-class data. In this case, a reasonable alternative for equilibrium capacity computation is provided by Theorem 1, which allows the manager to calculate explicitly its *best response* assignment, namely the capacity assignment that will satisfy the fixed ratio objective given the *current* network flows. This calculation requires just the total flows in each service class, or alternatively the delays thereof, which are easy to measure. Theorem 2 suggests that if the manager adopts its best response assignment and the network reaches an equilibrium, then this equilibrium is the unique NEP which satisfies the ratio objective. The analysis of such a dynamic scheme, however, is beyond the scope of this paper.

A possible criticism of the fixed ratio objective, as defined in (3), is that it does not account at all for absolute congestion measures, namely the delays themselves rather then their ratios. However, Theorem 3 shows that, by achieving the ratio objective, the manager in fact minimizes the cost function $\bar{J}^M$, which is just an appropriately weighted sum of the delays over the different service classes. Note that the weights β_a are inversely proportional in ρ_a^2, which gives higher weight to better (lower relative delay) service classes, as might be expected. From an algorithmic point of view, it should be noted that $\bar{J}^M$ is a convex and continuous function, and therefore may provide a sound basis for any iterative (gradient-based) algorithm that may be used by the manager to minimize this cost, with the goal of

eventually reaching the desired fixed-ratio equilibrium. Again, a specific consideration of such an algorithm is beyond the scope of this work.

We finally mention the notion of a Stackelberg equilibrium [11], which is sometimes used as an alternative to the Nash equilibrium in network analysis. In a Stackelberg game, the leading player (in our case the network manager) announces his strategy first, and the other players react to this strategy. Thus, the leading player should choose his strategy while taking into account the reaction of the others to it. However, as in our case the NEP leads to a global minimum (of 0) of the ratio objective, it may be seen that the Stackelberg equilibrium in fact coincides with the Nash equilibrium, and both are given by the results above.

3.3 proofs

Proof of Theorem 1: Observe that the best response allocation must meet $c_a - f_a > 0$ for every $a = 1, \ldots, A$, otherwise Assumption 1(ii) would obviously be violated. The best response allocation sets the ratios for the current flow distribution $(f_1, \ldots, f_A)$. It is derived from solving the following set of *linear* equations:

$$\sum_{a=1}^{A} c_a = C \tag{8}$$

$$\rho_a(c_a - f_a) = (c_1 - f_1) \quad \text{for every } a = 2, \ldots, A. \tag{9}$$

The unique solution of this set yields the capacities via

$$c_a = f_a + (C - \sum_{k \in \mathcal{A}} f_k) \frac{\rho_a^{-1}}{\sum_{k \in \mathcal{A}} \rho_k^{-1}}, \tag{10}$$

where $\rho_1 \stackrel{\triangle}{=} 1$. Thus, by solving (8)-(9) we get finite (positive) delays D_a, which meet the desired ratios, specified by the vector ρ. □

Proof of Theorem 3: The proof is based on characterizing the manager best response in the case of using $\bar{J}^M$. This characterization is established in the following lemma

Lemma 1 *Consider a fixed flow vector $(f_1, \ldots, f_A)$, and a manager whose cost function is given by (5). Then $\mathbf{c} = (c_1, \ldots, c_A)$ is the manager's best response to (5) if and only if the*

following relations hold for every $a, k \in \mathcal{A}$:

$$\sqrt{\beta_a} D_a(c_a, f_a) = \sqrt{\beta_k} D_k(c_k, f_k). \tag{11}$$

Proof: Observe first that the optimal capacity allocation must obey $0 < c_a < C$ for every $a = 1, \ldots, A$. Indeed, if $c_a = 0$ for some $a \in \mathcal{A}$ then the manager cost is infinite, as the summand $\beta_a D_a(f_a, 0)$ in (5) is infinite; if $c_a = C$ for some $a \in \mathcal{A}$ then the manager cost is also infinite, as all summands $\beta_k D_k(f_k, c_k)$ $k \in \mathcal{A}, k \neq a$ in (5) are infinite. Thus, the only active constraint of the manager is $\sum_{a \in \mathcal{A}} c_a = C$. Consequently, the KKT optimality conditions (which are necessary and sufficient in our case, since $\beta_a D_a(f_a, c_a)$ is convex in c_a) imply that $\frac{d\bar{J}^M}{dc_a} = \frac{d\bar{J}^M}{dc_k}$ for every $a, k \in \mathcal{A}$ (this equality is due to the equality of each $\frac{d\bar{J}^M}{dc_a}$ to the lagrange multiplier of the active constraint). Noting that $\frac{d\bar{J}^M}{dc_a} = \frac{-\beta_a}{(c_a - f_a)^2}$, we get

$$\frac{\beta_a}{(c_a - f_a)^2} = \frac{\beta_k}{(c_k - f_k)^2}. \tag{12}$$

Taking a square root from both sides of the last equation (and noting that $c_a - f_a > 0$ for every $a \in \mathcal{A}$) gives the required relations (11). □

Now substituting $\beta_a = \frac{1}{\rho_a^2}$ for every $a = 1, \ldots, A$, $\rho_1 \stackrel{\triangle}{=} 1$ in (11) yields the delay ratios for ρ. This ratios are also the manager best response when using J^M with ρ as the desired ratios. Thus the best responses of J^M with ρ and $\bar{J}^M$ with β coincide. Since the users always use (2) as their objective function, the equilibrium points of both users-manager games (with the relation between ρ and β as described above) coincide. □

Proof of Theorem 2: The existence and uniqueness proof is based on the existence and uniqueness of the users-manager game, where the manager uses $\bar{J}^P$. The existence and uniqueness of the latter game are established in the next two lemmata

Lemma 2 *Consider a users-manager game, in which the cost function of each user is given by (2) and the cost function of the manager is given by (5). This game has a Nash equilibrium point.*

Proof: Since $\bar{J}^M$ is convex in $\mathbf{c}$, J^i is convex in $\mathbf{f}^i$ $\forall i \in \mathcal{I}$, and the decision region of each user is convex, the existence of an equilibrium point is due to Rosen ([22], Theorem 1). One technical detail that has to be checked is that the costs of all game participants are finite in any NEP. As the total demand does not exceed the total capacity, this conclusion follows straightforwardly. Since the best response of either the manager or the users is always finite, in any situation where a cost function has an infinite value, either the manager or the user(s) with the infinite cost may react by modifying the capacity allocation or the routing decisions to encounter finite costs. The same proof idea was used in [20] for network routing games. □

Lemma 3 *The equilibrium point of the users-manager game, given by cost functions (2) and (5) is unique.*

Proof: See appendix.

The proof of the theorem may now be completed. Since the equilibrium point of the users-manager game with $\bar{J}^M$ exists and is unique, Theorem 3 implies that so is the case when J^M replaces $\bar{J}^M$. Hence, there is a unique equilibrium point where the ratios are met[†]. □

Proof of Theorem 4: The proof idea is that for any given ρ, the desired equilibrium delays $D_1, \ldots, D_A$ are essentially given (Lemma 4). We use this fact in order to calculate the equilibrium flows, and as a by-product, the desired capacity allocation. Our method is based on expressing the KKT optimality equations of every user with respect to its cost function (2) (Lemma 5).

Lemma 4 *Given a total capacity of C, a total flow r ($r < C$) and a desired delay ratio vector ρ, the desired equilibrium delays $D_1, \ldots, D_a$ are uniquely given by*

$$D_a = \frac{\rho_a \sum_{k \in \mathcal{A}} \rho_k^{-1}}{(C - \sum_{k \in \mathcal{A}} f_k)} \tag{13}$$

for every $a \in \mathcal{A}$, where $\rho_1 \stackrel{\triangle}{=} 1$

[†]The existence proof may also be obtained by a direct application of the Kakutani fixed point theorem. However, a constructive proof was preferred here.

Proof: Noting that the equilibrium delays are in particular induced by the manager's best response to the equilibrium flows, we may inverse both sides of equation (13) to obtain the equilibrium delays. Since $(C - \sum_{k\in\mathcal{A}} f_k) = C - r$ is a fixed quantity, the equilibrium delays are calculated even without knowing the equilibrium flows. □

Since (2) is convex in the decision vector $\mathbf{f}^i = (f_1^i, \ldots, f_A^i)$, the KKT optimality conditions of (2) constitute necessary and sufficient conditions for the flow to be at equilibrium. We summarize the KKT optimality conditions of the desired equilibrium point in the next lemma.

Lemma 5 *Let $D_1, \ldots, D_A$ be the (unique) delays for a ratio vector ρ. The following equations are met at the equilibrium point: For every $i \in \mathcal{I}$ there exist a (Lagrange multiplier) λ^i such that for every arc $a \in \mathcal{A}$*

$$\begin{aligned} D_a + f_a^i D_a^2 &\geq \lambda^i \quad \text{if } f_a^i = 0, \\ D_a + f_a^i D_a^2 &= \lambda^i \quad \text{if } 0 < f_a^i < s_a^i, \\ D_a + f_a^i D_a^2 &\leq \lambda^i \quad \text{if } f_a^i = s_a^i. \end{aligned} \tag{14}$$

Proof: Consider the equilibrium point where the ratios ρ are met (by Theorem 2 the equilibrium exists and is unique). It follows from Lemma 4 that the delays in the equilibrium point, namely $D_a = \frac{1}{c_a - f_a}, a = 1, \ldots, A$ are uniquely determined for every C, r and ρ. Define $J_a^i(f_a, c_a) \triangleq f_a^i D_a(f_a, c_a)$. Let $K_a^i(f_a^i, f_a, c_a) \triangleq \frac{dJ^i(\mathbf{f},\mathbf{c})}{df_a^i} = \frac{\partial J_a^i(f_a^i, f_a, c_a)}{\partial f_a^i} + \frac{\partial J_a^i(f_a^i, f_a, c_a)}{\partial f_a}$ denote the marginal costs of user i on arc a with respect to its flow f_a^i (where the second term is required since $f_a = \sum_i f_a^i$). We have

$$\begin{aligned} K_a^i(f_a^i, f_a, c_a) &= \frac{c_a - f_a^{-i}}{(c_a - f_a)^2} = \frac{1}{(c_a - f_a)} + \frac{f_a^i}{(c_a - f_a)^2} \\ &= D_a(f_a, c_a) + f_a^i D_a^2(f_a, c_a). \end{aligned} \tag{15}$$

We note that since the delays D_a are known at the equilibrium point (solved via (13)), the derivatives K_a^i in equilibrium can be written as a linear function of f_a^i only. By a slight abuse of notations we re-write equation (15), where $D_a(\cdot, \cdot)$ is replaced by a (given) constant D_a, expressing the derivatives K_a^i in equilibrium,

$$K_a^i(f_a^i) = D_a + f_a^i D_a^2. \tag{16}$$

Deriving the Lagrangian of (2) with respect to f_a^i, it follows that the KKT optimality equations of every user i are given by (14). □

We are now ready to prove the theorem. Consider the following quadratic optimization problem

$$\begin{cases} \min_{f_1^i,\ldots,f_A^i} \sum_{a=1}^{A} \frac{1}{2} D_a^2 {f_a^i}^2 + D_a f_a^i \\ \text{s.t} \qquad 0 \le f_a^i \le s_a^i \\ \qquad\quad \sum_{a=1}^{A} f_a^i = r^i \end{cases} \tag{17}$$

It is easy to see that the KKT optimality conditions of the above optimization problem coincide with the KKT conditions in (14), which are derived from the original user optimization problem. In other words, the unique solution vector $(f_1^i, \ldots, f_A^i)$ of (17) (as a quadratic problem) complies with the user KKT conditions, i.e., it is a user flow vector in the equilibrium point. Therefore, the equilibrium flow strategy $(f_1^i, \ldots, f_A^i)$ $\forall i \in \mathcal{I}$ is solvable via I *independent* optimization problems. Having solved (17) for every $i \in \mathcal{I}$, the manager capacities may be easily calculated through

$$c_a = f_a + \frac{1}{D_a}, \tag{18}$$

where $f_a = \sum_{i \in I} f_a^i$. □

Proof of Theorem 5: The following lemma assures that once the capacities are set, there cannot be multiple equilibria.

Lemma 6 *Consider a network of parallel arcs where the cost function of each user is given by (2) and the capacities $c_1, \ldots, c_A$ are static. In addition, the flow of each user in every arc is limited to a constant $s_a^i \ge 0$, i.e., $0 \le f_a^i \le s_a^i$ $\forall i, a$. In such a network the Nash equilibrium point is unique.*

Proof: See Appendix. □

Remark 1 *The first part of the last theorem (the calculation of the Nash equilibrium) provides an alternative proof to the uniqueness of the equilibrium where the delay ratios are met. Indeed, assume that for some i there exists an additional equilibrium strategy, $(\hat{f}_1^i, \ldots, \hat{f}_A^i) \neq (f_1^i, \ldots, f_A^i)$. Then by the sufficiency of the KKT conditions (14), $(\hat{f}_1^i, \ldots, \hat{f}_A^i)$ solves (17), contradicting the strict convexity property of the quadratic cost functions.*

4 General Networks

In this section we extend our results to the general network case, in which every user has its own source and destination, and a given route which leads from that source to the destination. We assume that the route is predetermined by some routing protocol, and is not a part of the user decisions. This assumption is reasonable, as Diffserv users would most likely determine only the assignments of their traffic to the different service classes (with the economic expenses involved), trusting the network manager for suitable routing decisions.

4.1 Model Definitions

A general network consists of a set of links $\mathcal{L} = \{1, \ldots, L\}$. As before, let $\mathcal{I} = \{1, 2, \ldots, I\}$ be the set of users, which share the network. We associate with each user i a route $R^i = (l^{i,1}, \ldots, l^{i,N^i})$, where N^i is the length of the route and $l^{i,k}$ is the kth link traversed by user i. Let $\mathcal{I}_l \subset \{1, 2, \ldots, I\}$ be the set of users for which $l \in R^i$. Each link l carries a set of service classes $\mathcal{A} = \{1, 2, \ldots, A\}$. Hence, each link is comprised of a set of A parallel arcs (see Figure 1). Additionally, each link contains a capacity C_l, which is to be divided between its service classes. Denoting by c_{la} the capacity assigned to class a in link l, a capacity assignment is feasible as long as (i) $c_{la} \geq 0 \ \forall l, a$, and (ii) $\sum_{a \in \mathcal{A}} c_{la} = C_l$. Denote by $\mathbf{c}_l = (c_{l1}, \ldots, c_{lA})$ the capacity allocation vector for link l. Let Γ_l denote the set of all feasible capacity allocations for link l, and let $\Gamma = \Gamma_1 \times \ldots \times \Gamma_L$ be the set of all feasible network capacity allocations $\mathbf{c} = (\mathbf{c}_1, \ldots, \mathbf{c}_L)$ ($\mathbf{c}$ can be viewed as matrix of dimension $L \times A$).

Each user $i \in \mathcal{I}$ has a fixed throughput demand, which is a constant r^i. As its route is fixed, the user's only decision is how to split its demand among the service classes. Using the same notations as in the single link case, $0 \leq f_a^i \leq s_a^i$, and $\sum_{a \in \mathcal{A}} f_a^i = r^i$. Denoting by $f_a^{i,k}$ the flow assigned to the kth link of user i in service class a, we have $f_a^i = f_a^{i,1} = \ldots = f_a^{i,N^i}$; in other words, once the user has determined the inter-class flow distribution, it remains constant along the path. Turning our attention to some link $l \in \mathcal{L}$, let f_{la} be the total flow in link l which is assigned to service-class a, we have $f_{la} = \sum_{i \in \mathcal{I}_l} f_a^i$ for every $l \in \mathcal{L}, a \in \mathcal{A}$. The vector $\mathbf{f}_l = (f_{l1}, \ldots, f_{lA})$ represent the intra-link flow distribution. Finally, a system configuration is feasible if it is composed of feasible flow configurations and feasible capacity allocations.

Following Kleinrock's independence approximation [4], we assume that the delay of every service class in every link follows that of an M/M/1 queue; thus the delay D_{la} of service class a at link l is given by

$$D_{la}(f_{la}, c_{la}) = \begin{cases} \frac{1}{c_{la} - f_{la}} & f_{la} < c_{la} \\ \infty & \text{otherwise} \end{cases} \tag{19}$$

Each user i minimizes its total delay, namely,

$$J^i = \sum_{l \in R^i} \sum_{a=1}^{A} f_a^i D_{la}(f_{la}, c_{la}), \tag{20}$$

where D_{la} is taken from (19). An equivalent formulation to (20) is

$$J^i = \sum_{a=1}^{A} f_a^i D_a^i, \tag{21}$$

where D_a^i is the end-to-end user i delay in service-class a, $D_a^i = \sum_{l \in R^i} D_{la}(f_{la}, c_{la})$. The objective of the provider remains to impose predetermined ratios between the delays of the service classes. Formally, given a ratio vector $\rho = (\rho_2, \ldots, \rho_A)$, the provider's goal is that the delays D_a^i, $a = 1, \ldots, A$ will obey

$$D_a^i = \rho_a D_1^i, \quad a = 2, \ldots, A \ \forall i \in \mathcal{I}. \tag{22}$$

(In terms of the cost function J^M, $J^M = 0$ if (22) is met; and ∞ otherwise). We also define the link-wise objective of minimizing the weighted sum of the delays, that is

$$\bar{J}^{M_l} = \sum_{a \in \mathcal{A}} \beta_a D_{la}(f_{la}, c_{la}), \tag{23}$$

where $\beta_a > 0 \ \forall a \in \mathcal{A}$.

Throughout this section, the following feasibility assumption holds:

Assumption 2

(i) $r^i \leq \sum_{a=1}^{A} s_a^i$ *for every* $i \in \mathcal{I}$.

(ii) $\sum_{i \in \mathcal{I}_l} r^i < C_l$ *for every* $l \in \mathcal{L}$

As in the single link model, we use the Nash equilibrium approach for the capacity allocation. Under this approach, two sub-approaches may be considered. The first consists of a single manager that minimizes J^M. For this case, the equilibrium definition of (6) applies too. The second sub-approach is a *distributed* approach, where the capacity adaptation is performed locally, at the link level. For this case, a feasible system configuration $(\tilde{\mathbf{f}}, \tilde{\mathbf{c}})$ is a NEP if the following conditions hold:

$$J^{M_l}(\tilde{\mathbf{f}}, \tilde{\mathbf{c}}_l, \tilde{\mathbf{c}}_{-l}) = \min_{\mathbf{c}_l \in \Gamma_l} J^{M_l}(\tilde{\mathbf{f}}, \mathbf{c}_l, \tilde{\mathbf{c}}_{-l}) \quad \forall l \in \mathcal{L}, \tag{24}$$

$$J^i(\tilde{\mathbf{f}}^i, \tilde{\mathbf{f}}^{-i}, \tilde{\mathbf{c}} = \min_{\mathbf{f}^i \in \mathbf{F}^i} J^i(\mathbf{f}^i, \ldots, \tilde{\mathbf{f}}^{-i}, \tilde{\mathbf{c}}) \quad \forall i \in \mathcal{I}.$$

We will refer to the interaction of the distributed approach as the *users-managers* game.

4.2 Main Results

In this section we suggest schemes for obtaining the ratios for general networks. We prove the existence and uniqueness of an equilibrium point in which the desired ratios are met (for every given ρ). Unlike the single link case, the equilibrium point is not necessarily unique, a fact that we demonstrate through an example. Accordingly, we focus on the equilibrium which preserves the delay ratios on a link-basis, and enjoys the added advantage that it may be induced by a distributed (link-based) management scheme. This equilibrium always exists and is unique. As before, we show the equivalence of the best response capacity assignment w.r.t. the ratio objective (22) to the best response w.r.t. the modified objective function (23). The last fact implies on appealing distributed algorithms, who preserve the ratios on a link-basis. Finally, we show that the equilibrium, which preserves the delay ratios on a link-basis, can be *explicitly calculated* via solving quadratic optimization problems. The complexity of computing the equilibrium remains the same as in the single link case. We summarize the results in the following theorems.

Theorem 6 *For every ρ, there exists an equilibrium point where the ρ ratios are met.*

Theorem 7 *The equilibrium point where the ρ ratios are met is not necessarily unique.*

Theorem 8 *Consider a users-managers game, where each manager cost function is given by (23), with the same β for all managers. Let ρ be a desired and ratio vector. Then, there*

exists an equilibrium point for the above game. Furthermore, if the β vector obeys $\beta_a = \frac{1}{\rho_a^2}$ for every $a = 1, \ldots, A$, $\rho_1 \stackrel{\triangle}{=} 1$, then the delay ratios ρ are met in any equilibrium of the game.

Theorem 9 *For every ratio vector ρ, there exists a unique equilibrium point in which the required ratios are preserved on a link-basis. Hence, the game described in Theorem 8 has a unique equilibrium point.*

Theorem 10 *The equilibrium point which preserves the ratio vector ρ on a link basis, can be explicitly calculated. This requires solving I quadratic optimization problems with A variables each.*

The implications of the these results are summarized in the following remarks

1. The significance of Theorems 8 and 9 is that the ratios can be kept by applying local capacity management. The only requirement for the adaptation is the local link delays. As in the single link case, the manager best response can be translated into a smooth adaptation process, based on convex cost functions. The minimization of these functions is still related to the social welfare.

2. The computation of the Nash equilibrium scales well to the general network case. Like in the single link case, we still solve I quadratic problems with A variables each. The only additional overhead is in calculating certain constants which are needed for the optimization problems. Nonetheless, these calculations impose insignificant computational overhead in comparison to solving the optimization problems themselves.

3. The ability to compute the unique Nash equilibrium where the ratios hold on a link-basis indicates that the manager can explicitly calculate the capacity allocation of that equilibrium. Nevertheless, one should be cautious when applying this approach in this case. At this point, it is not clear whether the user equilibrium flows (for a fixed capacity allocation) are unique for a network with general topology and the given cost functions ([20], [21]). In case that uniqueness does not hold, the setting of capacities via the computation approach should be accompanied by auxiliary management in order to lead the network to the desired equilibrium. Another alternative to be considered is the reactive (best-response) capacity assignment, as described in Section 3.2.

4.3 Proofs

Proof of Theorem 8: The proof of Lemma 2 (the existence of a NEP for the single link case) is easily extended to the general network case, as the user cost function remains convex in its flow vector, while each manager cost function is convex in its capacity allocation vector. Moreover, the condition $\sum_{i\in\mathcal{I}_l} r^i < C_l$ for every $l \in \mathcal{L}$ ensures that the best response of all game participants is finite. Thus the existence of a NEP is guaranteed as a convex game, by the same reasoning used in Lemma 2. We may now apply the arguments used in Theorem 3 to conclude that in any NEP $(\hat{\mathbf{f}}, \hat{\mathbf{c}})$ of the users-managers interaction (as defined in (24))

$$\sqrt{\beta_a} D_{la}(\hat{f}_{la}, \hat{c}_{la}) = \sqrt{\beta_k} D_{lk}(\hat{f}_{lk}, \hat{c}_{lk}) \ \forall a, k \in \mathcal{A}, l \in \mathcal{L},$$

or

$$D_{la}(\hat{f}_{la}, \hat{c}_{la}) = \rho_a D_{l1}(\hat{f}_{l1}, \hat{c}_{l1}) \ a = 2, \ldots, A, \forall l \in \mathcal{L}. \tag{25}$$

Since the ratios are maintained on a link-basis it follows that the end-to-end delay ratios of each user comply with the ratio vector ρ. □

Proof of Theorem 6: The proof follows directly from the proof to Theorem 8. In that proof, we show that for every ρ there exists an equilibrium point of the users-managers game, where the ρ delay ratios are met on a link-basis, and thus end-to-end. This equilibrium is obviously an equilibrium when the managers are replaced by a single manager M whose cost function is J^M. □

Proof of Theorem 10: As in the single link case, the network delays for the chosen equilibrium are known and uniquely determined. Based on the KKT optimality conditions of the users, we are able to define quadratic optimization problem for each user; from the solutions to the optimization problems, we obtain the needed capacities for the equilibrium.

Formally, consider an equilibrium point where the ratios ρ are met on a link-basis (by Theorem 6 there is such an equilibrium). It follows from Lemma 4 that the delays in that equilibrium point, namely D_{la} $\forall l, a$, are uniquely determined (note that the total demand f_l in each link is constant, as each user has a single route). We may therefore follow the reasoning in the proof of Lemma 5, and express the marginal (path-wise) cost $K_a^i \triangleq \frac{dJ^i(\mathbf{f},\mathbf{c})}{df_a^i}$ as a function of f_a^i and the delays along the path:

$$K_a^i(f_a^i) = \sum_{l\in R^i} D_{la} + f_a^i D_{la}^2 = \sum_{l\in R^i} D_{la} + f_a^i \sum_{l\in R^i} D_{la}^2. \tag{26}$$

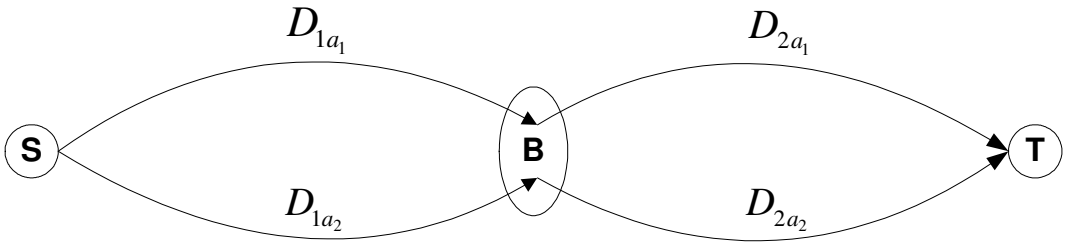


Figure 1: A network with a source S, an intermediate node B and a destination T. There are two links, S–B and B–T. In this network $A = 2$, thus each link contains two parallel arcs. We emphasize in the drawing that flows do not switch from one service class to the other.

Denoting $D_a^{i\,2} \triangleq \sum_{l \in R^i} D_{la}^2$, the KKT optimality equations of every user i can be written as

$$\begin{aligned} D_a^i + f_a^i D_a^{i\,2} &\geq \lambda^i \quad \text{if } f_a^i = 0, \\ D_a^i + f_a^i D_a^{i\,2} &= \lambda^i \quad \text{if } 0 < f_a^i < s_a^i, \\ D_a^i + f_a^i D_a^{i\,2} &\leq \lambda^i \quad \text{if } f_a^i = s_a^i. \end{aligned} \tag{27}$$

We may therefore define the same quadratic optimization problem (17) (with D_a^i replacing D_a, and $D_a^{i\,2}$ replacing D_a^2) for every $i \in \mathcal{I}$ to obtain the equilibrium flows for each user. Having all flows at hand the capacities are

$$c_{la} = f_{la} + \frac{1}{D_{la}}, \tag{28}$$

for every $l \in \mathcal{L}$ and $a \in \mathcal{A}$. □

Proof of Theorem 9: Based on the proof of Theorem 10, the proof to Theorem 9 is now straightforward. As the equilibrium flows are obtained by solving strictly convex optimization problems, the equilibrium flows are uniquely determined; so is the capacity allocation. In other words, there exists a unique equilibrium point in which the required ratios are preserved on a link-basis. □

Proof of Theorem 7: The following example shows that an equilibrium with the desired ratios is generally not unique. This fact can be intuitively understood, as the general network architecture provides additional degrees of freedom for the capacity setting in comparison to the single link case. Consider the network depicted in Figure 1. It contains a source node S, a destination node T and an intermediate node B. Thus all users have the same route

S–B–T. The link between S and B (indexed as 1) is equipped with a total capacity of C_1, while the link between B and T (indexed as 2) is equipped with a total capacity of C_2. The network offers two service classes a_1 and a_2, which should provide an end-to-end delay ratio of ρ_2, i.e., $D_2^i = \rho_2 D_1^i$, $i = 1, \ldots, I$. The capacity allocation which was selected in Theorem 10 forces the delay ratios on a link basis. Yet, we may force a different delay ratio between the arcs of link 1 (say $\gamma_2 \neq \rho_2$), and then compensate for this deviation in the capacity allocation of link 2, as follows: Once γ_2 is set, D_{1a_1}, D_{1a_2} are known (by Lemma 4); we are left with two variables D_{2a_1}, D_{2a_2} which are solved from the following equation set:

$$\begin{cases} D_{1a_2} + D_{2a_2} = \rho_2 (D_{1a_1} + D_{2a_1}) \\ \frac{1}{D_{2a_1}} + \frac{1}{D_{2a_2}} = C_2 - r, \end{cases} \tag{29}$$

where r is the users' total demand. In order to further illustrate this principle, we provide a numerical example for the network, with two different equilibria which yield the same end-to-end delay ratios.

There are two users with demands of $r^1 = 4.5, r^2 = 4$, with no restricting SLAs (i.e., $s_a^i \geq r^i$ for every i, a); in addition $C_1 = C_2 = 10$, $\rho_2 = 2$. In the first equilibrium we force the delay ratio $\rho_2 = 2$ at every link, yielding delays of $D_{1a_1} = D_{2a_1} = 1$, $D_{1a_2} = D_{2a_2} = 2$ (see the proof of Lemma 4 for the calculation). In the second equilibrium we force a delay ratio of $\gamma_2 = 3$ (and thus $D_{1a_1} = \frac{8}{9}$, $D_{1a_2} = \frac{8}{3}$), and by (29) we obtain $D_{2a_1} = 1.20, D_{2a_2} = 1.50$ (results rounded up to the second digit after the point). Interestingly, the equilibrium flows can be calculated (in either equilibrium) by solving a simple linear equation set (and not the quadratic optimization problems (17)); this is true since in congested situations (where the total flow is close to the total link capacity), users use all service classes. Thus the KKT conditions of (27) sum up to a second linear equation necessary for each user (the first equation is the total demand equation):

$$D_{a_1}^i + f_{a_1}^i D_{a_1}^{i^2} = D_{a_2}^i + f_{a_2}^i D_{a_2}^{i^2}. \tag{30}$$

Finally, the capacities are calculated via (28). To summarize, the flows and capacities of the first equilibrium are $\big(f_{a_1}^1 = 3.8, f_{a_2}^1 = 0.7, f_{a_1}^2 = 3.4, f_{a_2}^2 = 0.6, c_{1a_1} = 8.2, c_{1a_2} = 1.8, c_{2a_1} = 8.2, c_{2a_2} = 1.8\big)$, and those of the second equilibrium are $\big(f_{a_1}^1 = 3.82, f_{a_2}^1 = 0.68, f_{a_1}^2 = 3.41, f_{a_2}^2 = 0.59, c_{1a_1} = 8.36, c_{1a_2} = 1.64, c_{2a_1} = 8.07, c_{2a_2} = 1.93\big)$. We observe that even the induced flows are not the same in both equilibria.

5 Pricing and Dynamic SLAs

In this section, we briefly outline how the principles of this work apply also for the dynamic SLAs case. Under the dynamic SLAs framework [24], a user buys the service *on-demand*, usually by paying a given price. Such possibility is favored by users who do not need enhanced QoS on a constant basis. Although the dynamic SLAs framework can be supported in parallel with static SLAs, let us assume, for the sake of exposition, that the network now offers only dynamic SLAs.

Assume each service class a is advertised with a constant price p_a. The prices are set by the network provider, and reflect certain economic considerations. The delay ratios ρ are also declared, and must be preserved. Concentrating on the single link network, the cost function of user i is now

$$J^i(\mathbf{f}, \mathbf{c}) = \sum_{a=1}^{A} f_a^i \left(p_a + \nu^i D_a(f_a, c_a)\right), \tag{31}$$

where $\nu^i > 0$ is a user-specific parameter, representing the price-quality tradeoff. We note that the convexity properties that were needed for our results to hold, still apply when (31) replaces the original cost functions. Specifically, the derivative of J^i with respect to f_a^i adds just a constant p_a, which does not affect the ability to define quadratic optimization problems for calculating the equilibrium with the desired ratios. These observations hold also for the network case.

We should take into account, however, that the full knowledge which is required for calculating the NEP includes the ν^i's; these can only be estimated. This suggests a possible combination between the computation approach, which may perform an initial capacity allocation based on rough estimates of the user's preferences, followed by an adaptive algorithm. Obviously, the last idea applies also for the static SLAs case, whenever only partial information is available.

6 Conclusion

In this paper, we considered an approach to QoS provisioning in Diffserv networks, which focuses on maintaining relative congestion measures in the different service classes. Our anal-

ysis demonstrated the feasibility of this approach. Our congestion model uses the common M/M/1 delay function, and furthermore takes into account a reactive and heterogenous user environment. An attractive feature of the suggested capacity allocation schemes is in the ability to implement per-link distributed algorithms. This also leads to efficient and feasible computation schemes for capacity assignment on a per link basis.

Additional research is called for to enhance the scope of our model and analysis. Other congestion measures should be considered, starting perhaps with the addition of propagation delay measures to the link delays, and proceeding to the explicit incorporation of other QoS measures. The study of *equilibrium dynamics*, namely iterative or incremental capacity allocation algorithms and their convergence properties, is of obvious interest. We shall examine coupled cost functions, where the performance of each service class may depend on the congestion level at other classes. These may provide better models for actual scheduling schemes such as the GPS and the Weighted Round Robin schedulers.

A Appendix

We complete the proofs that were omitted from the main text. In fact, these proofs extend the results of [20] regarding the uniqueness of the Nash equilibrium for the parallel arcs network. We stick to the terminology and notations that where used in [20]. Let us first restate the type-A cost functions properties that were used in [20]:

A1 J^i is the sum of arc cost functions i.e., $J^i(\mathbf{f}) = \sum_{a\in\mathcal{A}} J^i_a(\mathbf{f}_a)$. Each J^i_a satisfies:

A2 $J^i_a : [0,\infty)^I \to [0,\infty]$, a continuous function.

A3 J^i is convex in f^i_a.

A4 Wherever finite, J^i_a is continuously differentiable in f^i_a. We denote: $K^i_a = \frac{\partial J^i_a}{\partial f^i_a}$.

A5 For every flow configuration $\mathbf{f}$, if not all user costs are finite then at least one user with infinite cost ($J^i(\mathbf{f}) = \infty$) can change its own flow configuration to make its cost finite.

A6 J^i_a is a function of two arguments, namely user i's flow on arc a and the total flow on that arc. In other words: $J^i_a(\mathbf{f}_a) = \bar{J}^i_a(f^i_a, f_a)$

A7 $\bar{J}_a^i$ is increasing in each of its two arguments. By an abuse of notations, we shall denote the cost function as $J_a^i(f_a^i, f_a)$ (instead of $\bar{J}_a^i(f_a^i, f_a)$).

A8 Note that $K_a^i = K_a^i(f_a^i, f_a)$ is now a function of two arguments. We assume that whenever J_a^i is finite, K_a^i is strictly increasing in each of its two arguments.

We introduce a new type of cost functions, namely the type-$\tilde{A}$ cost functions. These functions follow properties A1-A5, where A6-A8 are replaced by the following assumptions:

$\tilde{A}$6 J_a^i is a function of the total flow on that arc. In other words: $J_a^i(\mathbf{f}_a) = \bar{J}_a^i(f_a)$

$\tilde{A}$7 $\bar{J}_a^i$ is increasing in its argument. By an abuse of notations, we shall denote the cost function as $J_a^i(f_a)$ (instead of $\bar{J}_a^i(f_a)$).

$\tilde{A}$8 Note that $K_a^i = K_a^i(f_a)$ is now a function of f_a only. We assume that whenever J_a^i is finite, K_a^i is strictly increasing in f_a.

Proof of Lemma 6: We prove the lemma for general type-A cost functions. The user cost function (2) is a special case of type-A cost functions. Let us restate the lemma for the general setting:

Consider a network of parallel arcs where the cost function of each user is of type-A. In addition, the flow of each user in every arc is limited to a constant $s_a^i \geq 0$, i.e., $0 \leq f_a^i \leq s_a^i \ \forall i, a$. In such a network the Nash equilibrium point is unique.

Proof: Let $\mathbf{f} \in F$ and $\hat{\mathbf{f}} \in F$ be two NEPs. $\mathbf{f}$ (and $\hat{\mathbf{f}}$) satisfy the KKT conditions, which may be written as

$$K_a^i(f_a^i, f_a) \ \geq \ \lambda^i \quad \text{if } f_a^i = 0, \tag{32}$$

$$K_a^i(f_a^i, f_a) \ = \ \lambda^i \quad \text{if } 0 < f_a^i < s_a^i, \tag{33}$$

$$K_a^i(f_a^i, f_a) \ \leq \ \lambda^i \quad \text{if } f_a^i = s_a^i. \tag{34}$$

The first step is to establish that $f_a = \hat{f}_a \ \forall a \in \mathcal{A}$. To this end, we prove that for each a and i, the following relations hold:

$$\left\{\hat{\lambda}^i \leq \lambda^i, \hat{f}_a \geq f_a\right\} \ \rightarrow \ \hat{f}_a^i \leq f_a^i, \tag{35}$$

$$\left\{\hat{\lambda}^i \geq \lambda^i, \hat{f}_a \leq f_a\right\} \;\; \rightarrow \;\; \hat{f}_a^i \geq f_a^i. \tag{36}$$

We shall only prove (35), since (36) is symmetric. Assume that $\hat{\lambda}^i \leq \lambda^i$ and $\hat{f}_a \geq f_a$ for some a and i. Note that (35) holds trivially if $\hat{f}_a^i = 0$ *or* if $f_a^i = s_a^i$, thus we only have to consider the case where $\hat{f}_a^i > 0$ and $f_a^i < s_a^i$. In that case, the KKT conditions (32) along with our assumptions imply that

$$K_a^i(\hat{f}_a^i, \hat{f}_a) \leq \hat{\lambda}^i \leq \lambda^i \leq K_a^i(f_a^i, f_a) \leq K_a^i(f_a^i, \hat{f}_a), \tag{37}$$

where the first and third inequalities follow from the KKT conditions, and the last inequality follows from the monotonicity of K_a^i in its second argument. Now, since K_a^i is strictly increasing in its first argument, this implies that $\hat{f}_a^i \leq f_a^i$, and (35) is established. Having (35) and (36) at hand, the exact arguments of Theorem 2.1 in [20] may be applied to show that

$$\hat{f}_a = f_a \quad \text{for every } a \in \mathcal{A}. \tag{38}$$

We shall repeat these arguments for the sake of completeness of the proof. Let $\mathcal{A}_1 = \left\{a : \hat{f}_a > f_a\right\}$, also denote $\mathcal{I}_a = \left\{i : \hat{\lambda}^i > \lambda^i\right\}$, $\mathcal{A}_2 = \mathcal{A} - \mathcal{A}_1 = \left\{a : \hat{f}_a \leq f_a\right\}$. Assume that $\mathcal{A}_1$ is not empty. Recalling that $\sum_{a\in\mathcal{A}} \hat{f}_a^i = \sum_{a\in\mathcal{A}} f_a^i = r^i$, it follows from (36) that for every $i \in \mathcal{I}_a$

$$\sum_{a\in\mathcal{A}_1} \hat{f}_a^i = r^i - \sum_{a\in\mathcal{A}_2} \hat{f}_a^i \leq r^i - \sum_{a\in\mathcal{A}_2} f_a^i = \sum_{a\in\mathcal{A}_1} f_a^i. \tag{39}$$

Noting that (35) implies that $\hat{f}_a^i \leq f_a^i$ for $a \in \mathcal{A}_1$ and $i \notin \mathcal{I}_a$, it follows that

$$\sum_{a\in\mathcal{A}_1} \hat{f}_a = \sum_{a\in\mathcal{A}_1} \sum_{i\in\mathcal{I}} \hat{f}_a^i \leq \sum_{a\in\mathcal{A}_1} \sum_{i\in\mathcal{I}} f_a^i = \sum_{a\in\mathcal{A}_1} f_a. \tag{40}$$

This inequality contradicts the definition of $\mathcal{A}_1$, which implies that $\mathcal{A}_1$ is an empty set. By symmetry it may also be concluded that the set $\left\{a : \hat{f}_a > f_a\right\}$ is empty. Thus, $\hat{f}_a = f_a$, $\forall a \in \mathcal{A}$. We now proceed to show that $\hat{f}_a^i = f_a^i \; \forall i, a$. To this end, note that (35) may be strengthened as follows:

$$\begin{aligned} &\left\{\hat{\lambda}^i < \lambda^i, \hat{f}_a = f_a\right\} \Rightarrow \\ &\hat{f}_a^i < f_a^i \quad \text{or} \quad \hat{f}_a^i = f_a^i = 0 \quad \text{or} \quad \hat{f}_a^i = f_a^i = s_a^i. \end{aligned} \tag{41}$$

Indeed, if $\hat{f}_a^i = 0$ the implication is trivial. So is the case where $f_a^i = s_a^i$. Otherwise, if $\hat{f}_a^i > 0$ and $f_a^i < s_a^i$ it follows similarly to (37) that $K_a^i(\hat{f}_a^i, \hat{f}_a) < K_a^i(f_a^i, \hat{f}_a)$, so that $\hat{f}_a^i < f_a^i$ as required. Using (41) we will contradict the event $\left\{\hat{\lambda}^i < \lambda^i, \hat{f}_a^i \neq f_a^i \text{ for some } a \in \mathcal{A}\right\}$. Following (38) and (41) this event is possible only if $\hat{f}_k^i < f_k^i$ for some $k \in \mathcal{A}$. But since by (41) $\hat{f}_a^i \leq f_a^i$, we will get a demand constraint violation, i.e., $\sum_{a\in\mathcal{A}} \hat{f}_a^i < \sum_{a\in\mathcal{A}} f_a^i$. Thus the only possibility to have $\hat{\lambda}^i < \lambda^i$ for some user $i \in \mathcal{I}$ is when $\hat{f}_a^i = f_a^i = s_a^i$ or $\hat{f}_a^i = f_a^i = s_a^i = 0$ for every $a \in \mathcal{A}$, i.e. $\hat{f}_a^i = f_a^i$ for every $a \in \mathcal{A}$. A symmetrical argument may be obtained for the case where $\hat{\lambda}^i > \lambda^i$. In other words, we showed that $\hat{f}_a^i = f_a^i \; \forall a \in \mathcal{A}$ for every user i such that $\hat{\lambda}^i \neq \lambda^i$. We conclude the proof by observing that for the remaining users with $\hat{\lambda}^i = \lambda^i$, a straightforward use of (35), (36) and (38) implies that $\hat{f}_a^i = f_a^i \; \forall a \in \mathcal{A}$.

□

Proof for Lemma 3: The main idea of the uniqueness proof is to view the manager as an additional user who submits flow into the network. Recall that the user's cost function is given by

$$J^i(\mathbf{f}, \mathbf{c}) = \sum_{a=1}^{A} f_a^i D_a(f_a, c_a), \tag{42}$$

where D_a is taken from (1), and the manager's cost function is given by

$$\bar{J}^M(\mathbf{f}, \mathbf{c}) = \sum_{a\in\mathcal{A}} \beta_a D_a(f_a, c_a), \tag{43}$$

1. The derivatives of the cost functions with respect to the decision variables are:

$$\frac{dJ^i}{df_a^i} = \frac{c_a - f_a + f_a^i}{(c_a - f_a)^2}, \tag{44}$$

$$\frac{d\bar{J}^M}{dc_a} = \frac{-\beta_a}{(c_a - f_a)^2}. \tag{45}$$

2. Define

$$f_a^M \triangleq C - c_a, \tag{46}$$

where C is the total capacity, and c_a is the capacity allocated to arc a. Under the above transformation, $\mathbf{f}^M = (f_1^M, \ldots, f_A^M)$ will be referred to as the manager decision vector.

Note that the manager feasibility constraints, $c_a \geq 0$ and $\sum_{a\in\mathcal{A}} c_a = C$, are now written as $f_a^M \leq C \ \forall a \in \mathcal{A}$ and $\sum_{a\in\mathcal{A}} f_a^M = (A-1)C$. From these two conditions it follows that $f_a^M \geq 0$ too, i.e.,

$$0 \leq f_a^M \leq C \text{ for every } a \in \mathcal{A}, \tag{47}$$

$$\textstyle\sum_{a\in\mathcal{A}} f_a^M = (A-1)C \stackrel{\triangle}{=} r^M. \tag{48}$$

3. Using the change of variables (46) in the derivatives (44)-(45) we obtain

$$\frac{dJ^i}{df_a^i} = \frac{C - f_a^M - f_a + f_a^i}{(C - f_a^M - f_a)^2}, \tag{49}$$

$$\frac{d\bar{J}^M}{df_a^M} = \frac{d\bar{J}^M}{dc_a}\frac{dc_a}{df_a^M} = \frac{\beta_a}{(C - f_a^M - f_a)^2}. \tag{50}$$

4. Denoting $\tilde{f}_a \stackrel{\triangle}{=} f_a + f_a^M$ we have

$$\frac{dJ^i}{df_a^i} = \frac{C - \tilde{f}_a + f_a^i}{(C - \tilde{f}_a)^2}, \tag{51}$$

$$\frac{d\bar{J}^M}{df_a^M} = \frac{\beta_a}{(C - \tilde{f}_a)^2}. \tag{52}$$

5. It follows from (47) that the manager may be regarded as an additional user with flow bounds on every arc and a total fixed "demand" r^M. Moreover, we observe that the feasibility constraint under the "transformed" game, $AC > \sum_{i=1}^I r^i + r^M$ obviously holds if the original feasibility constraint $C > \sum_{i=1}^I r^i$ is met. We notice from (51) that the user cost function is of type-A, and from (52) that the manager cost function is of type-$\tilde{A}$, with $\tilde{f}_a$ replacing f_a in both cases.

6. Straightforward application of Lemma 7 below yields that the equilibrium point is unique.

□

For the completeness of the proof the following Lemma is required:

Lemma 7 *Consider a network of parallel arcs where the cost functions of $I > 0$ users are of type-A and the cost function of an additional single user j is of type-$\tilde{A}$. In addition, the flow of every user in each arc is limited by a constant s_a^i, i.e., $0 \leq f_a^i \leq s_a^i \ \forall i, a$. In such a network the Nash equilibrium point is unique.*

Proof: Let $\mathbf{f} \in F$ and $\hat{\mathbf{f}} \in F$ be two NEPs. We first show that $\hat{f}_a = f_a$ for every $a \in \mathcal{A}$. Assume that the set $\mathcal{A}_1 = \left\{a : \hat{f}_a > f_a\right\}$ is not empty. That means that the set $\mathcal{A}_3 = \left\{a : \hat{f}_a < f_a\right\}$ is also not empty, in order not to violate the total demand constraint. Let us analyze the flow of the $I-1$ users whose cost functions are of type-A. Denote $\mathcal{I}_b = \left\{i : \hat{\lambda}^i > \lambda^i, i \neq j\right\}$, $\mathcal{A}_2 = \mathcal{A} - \mathcal{A}_1 = \left\{a : \hat{f}_a \leq f_a\right\}$. Recalling that $\sum_{a\in\mathcal{A}} \hat{f}_a^i = \sum_{a\in\mathcal{A}} f_a^i = r^i$, it follows from (36) that for every $i \in \mathcal{I}_b$

$$\sum_{a\in\mathcal{A}_1} \hat{f}_a^i = r^i - \sum_{a\in\mathcal{A}_2} \hat{f}_a^i \leq r^i - \sum_{a\in\mathcal{A}_2} f_a^i = \sum_{a\in\mathcal{A}_1} f_a^i. \tag{53}$$

Noting that (35) implies that $\hat{f}_a^i \leq f_a^i$ for $a \in \mathcal{A}_1$ and $i \notin \mathcal{I}_b \cup \{j\}$, it follows that

$$\sum_{a\in\mathcal{A}_1} \sum_{i\in\mathcal{I}\setminus\{j\}} \hat{f}_a^i \leq \sum_{a\in\mathcal{A}_1} \sum_{i\in\mathcal{I}\setminus\{j\}} f_a^i. \tag{54}$$

This inequality implies that if $\mathcal{A}_1 = \left\{a : \hat{f}_a > f_a\right\}$ is not empty set, user j must have $\hat{f}_a^j > f_a^j$ for at least one arc $a \in \mathcal{A}_1$. A symmetric argument for the set $\mathcal{A}_3$ shows that j must have $\hat{f}_a^j < f_a^j$ for at least one arc $a \in \mathcal{A}_3$. We will show that the above requirements on the flow of user j may not hold simultaneously. To that end, we prove the following relations

$$\left\{\hat{f}_a > f_a, \hat{f}_a^j > f_a^j\right\} \ \Rightarrow \ \hat{\lambda}^j > \lambda^j, \tag{55}$$

$$\left\{\hat{f}_a < f_a, \hat{f}_a^j < f_a^j\right\} \ \Rightarrow \ \hat{\lambda}^j < \lambda^j. \tag{56}$$

We will only prove (55) since (56) is symmetric. Note that since $\hat{f}_a^j > f_a^j$ we have $\hat{f}_a^j > 0$ and $f_a^j < s_a^j$. Let us assume that $\hat{\lambda}^j \leq \lambda^j$. Then we have

$$K_a^j(\hat{f}_a) \leq \hat{\lambda}^j \leq \lambda^j \leq K_a^j(f_a) < K_a^j(\hat{f}_a), \tag{57}$$

where the first and third inequalities follow directly from the KKT conditions, and the last (strict) inequality from the monotonicity of K_a^j in f_a. The inequality in (57) is obviously a contradiction, implying that $\hat{\lambda}^j > \lambda^j$.

Now since $\mathcal{A}_1 \neq \emptyset$ we conclude by (54) and (55) that $\hat{\lambda}^j > \lambda^j$. Similarly, since $\mathcal{A}_3 \neq \emptyset$ we must have $\hat{\lambda}^j < \lambda^j$, reaching a contradiction. This implies that both $\mathcal{A}_1$ and $\mathcal{A}_3$ are empty sets, i.e., $\hat{f}_a = f_a$ for every $a \in \mathcal{A}$.

Since $\hat{f}_a = f_a$ for every $a \in \mathcal{A}$, we may use (41) and the arguments thereafter to conclude that $\hat{f}_a^i = f_a^i$ for every $i \in \mathcal{I}\backslash\{j\}$ and $a \in \mathcal{A}$. Observing that

$$\hat{f}_a^j = \hat{f}_a - \sum_{\mathcal{I}\backslash\{j\}} \hat{f}_a^i = f_a - \sum_{\mathcal{I}\backslash\{j\}} f_a^i = f_a^j \quad \text{for every } a \in \mathcal{A}, \tag{58}$$

finalizes the proof of uniqueness.

□

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